AN ENVELOPE SEPARATION AIDED MULTI-TASK LEARNING MODEL FOR BLIND SOURCE COUNTING AND LOCALIZATION

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ABSTRACT

Sound source localization (SSL) under unknown or variable sound sources conditions remains a challenging task. Existing methods suffer from limitations such as grid resolution constraints, fixed output dimensionality and insufficient exploitation of mutual assitance between temporal and spatial information. In this paper, we propose an **E**nvelope **S**eparation **A**ided **M**ulti-**T**ask **L**earning model for blind source counting and localization, which adaptively generates attractors to estimate source numbers and jointly optimizes envelope separation and direction estimation through a multi-task learning model using permutation invariant training (PIT). Experimental results demonstrated that the proposed model achieved better performance, by leveraging temporal domain envelope separation to aid spatial localization, outperforming baseline approaches.

Index Terms— Sound source localization, direction of arrival, multi-task learning model, blind source counting

1. INTRODUCTION

Sound source localization (SSL) aims to estimate the direction of arrival (DOA) and distance of one or more sound sources, with applications in audio signal enhancement, robotics and smart home interaction [1]. Despite recent advances, accurate localization remains challenging, especially when the number of sources is unknown or variable. Conventional approaches including TDOA [2], MUSIC [3]), and beamforming [4] are highly sensitive to noise or reverberation, and rely on prior knowledge of acoustic parameters. Deep learning-based methods have improved robustness and can be broadly categorized into classification-based and regression-based approaches. Classification-based methods treat SSL as a classification problem, predicting the most likely direction from a predefined grid (such as azimuth-elevation bins) by mapping acoustic features to spatial labels [5]. However, the resolution of classification-based approaches is constrained by grid granularity and suffers from permutation ambiguity under multi-source conditions [6]. In contrast, regression-based SSL methods directly map acoustic features to continuous coordinates (such as Cartesian or Spherical) [7], but they generally assumes a fixed output dimensionality, restricting their applicability to blind source localization.

To address the problem of blind source counting and localization, several methods have been proposed, which can be broadly categorized into two groups: one employs thresholding mechanisms to estimate the presence probability of sources, while the other utilizes attractor-based deep neural networks to distinguish different sources in an embedding space. Within the first category, [8] introduced a CRNN-based framework for joint sound event detection (SED) and DOA estimation, enabling localization without prior knowledge of the source numbers. However, this approach struggles with homogeneous overlapping sounds. Track-wise methods such as EINV2 [9] improved upon this by employing per-track outputs with permutation-invariant training (PIT) [10] to predict individual sound events and DOAs, offering better handling of overlapping sounds. The ACCDOA method [11] integrates both SED and DOA estimation tasks into a unified representation using Cartesian DOA vectors, and mACCDOA [12], further enhances this, using duplicated PIT for overlapping sound scenarios. Although these methods address blind source counting, their outputs remain fixed-dimensional, and source existence is determined via empirically set thresholds, which can reduce accuracy in multi-source scenarios. In the second category, attractor-based network [13][14] has shown promise for handling unknown source numbers in tasks such as speech separation and diarization. Subsequent studies have combined attractor networks with Transformer architectures to achieve higher-quality separation and speaker binarization [15][16]. Specifically, [17] proposed SEET, a deep attractor-based method that estimates both the number of sources and their temporal envelopes, subsequently using envelopes as clues to guide target source localization. However, existing attractor-based methods generally perform source or envelope separation first, followed by localization as a separate step. They treat envelope estimation and DOA prediction as independent and sequential tasks, which reduces separation accuracy in multi-source scenarios and degrades subsequent localization performance.

The term "attractor" originates from Perceptual Magnet Effect [18], which suggests that biological neural networks form perceptual attractors that draw acoustically similar sounds closer together. Neurophysiological studies further indicate that human auditory system integrates temporal and spatial cues to improve SSL accuracy [19][20], implying that envelope separation and DOA prediction can mutually assist each other in SSL tasks. In this paper, we propose an **E**nvelope **S**eparation **A**ided **M**ulti-**T**ask **L**earning model that adaptively generates attractors to estimate source numbers and jointly optimizes envelope separation and SSL using PIT[10]. The main contributions are: (1) a multi-task learning model combining envelope and direction prediction to improve localization; (2) an attractor-based framework capable of handling blind source counting and localization. The rest of the paper is organized as follows: Section 2 describes the framework of proposed model, Section 3 presents experimental results, and Section 4 concludes the paper.

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2. METHOD

The overall framework of the proposed model is illustrated in Fig. 1. The model takes mixed First-Order Ambisonics (FOA) signals as input and simultaneously outputs four predictions for each active source: existence probability, DOA, distance and envelope, without relying on networks with fixed-dimensional outputs. The framework consists of three modules: the ***Acoustic Feature Extraction Module (AFEM)*** encodes FOA signals into time-frequency domain embeddings; the ***Adaptive Attractor Module (AAM)*** generates a dynamic attractor for each sound source along with source existence probabilities; and the ***Multi-Task Learning Module (MTLM)*** jointly estimates DOA, distance and envelope through a multi-task learning framework using PIT [10].

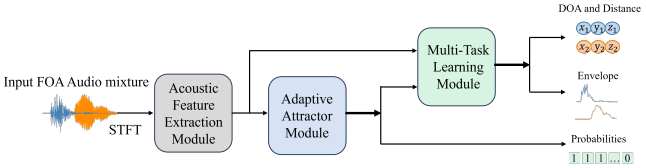
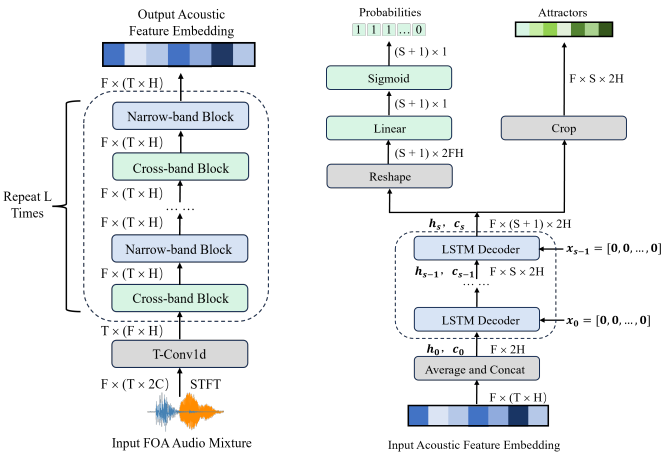


Fig. 1: The overall framework of proposed model.

2.1. Acoustic Feature Extraction Module

The Acoustic Feature Extraction Module (AFEM) is inspired by the encoder structure of SpatialNet [21], as shown in Fig. 2a. The mixed FOA signal is first transformed to the time-frequency domain using Short-Time Fourier Transform (STFT). A time-domain convolution extracts initial features, followed by Cross-band Blocks extracting inter-band features and Narrow-band Blocks extracting temporal features. The output acoustic feature embedding of the mixed signal is represented as $\mathbf{B}_{mix} \in \mathbb{R}^{F\times T\times H}$, where F, T, and H denote the number of frequency bins, time domain frames, and feature dimension, respectively.



(a) AFEM. (b) AAM.

Fig. 2: The structure of modules: (a) Acoustic Feature Extraction Module; (b) Adaptive Attractor Module.

2.2. Adaptive Attractor Module

The Adaptive Attractor Module (AAM), illustrated in Fig. 2b, addresses the challenge of unknown and variable sound source numbers by generating a flexible set of attractors. Following the LSTM-based decoder in [17], the AFEM output $\mathbf{B}_{mix}$ is averaged over the time domain dimension and concatenated along the feature dimension, generating $\mathbf{B}_{avg} \in \mathbb{R}^{F\times 2H}$. This representation initializes the hidden and cell states $(\mathbf{h}_0, \mathbf{c}_0)$ of a unidirectional LSTM, while a zero vector $\mathbf{x}_0 \in \mathbb{R}^{(S+1)\times 2H}$ is input at each decoding step. The hidden state $\mathbf{h}_{s-1}$, cell state $\mathbf{c}_{s-1}$, and attractor $\mathbf{a}_{s-1}$ for the s-th source are updated according to:

$$\mathbf{h}_{s-1}, \mathbf{c}_{s-1}, \mathbf{a}_{s-1} = \mathbf{LSTM}(\mathbf{h}_{s-2}, \mathbf{c}_{s-2}, \mathbf{0}), \tag{1}$$

where $\mathbf{a}_{s-1} \in \mathbb{R}^{F\times 1\times H}$ denotes the attractor vector for source s. Each attractor $\mathbf{a}_i$ is reshaped and concatenated into $\mathbb{R}^{1\times 2FH}$ and passed through a fully connected layer with sigmoid activation to estimate the probability $P_{pred} \in \mathbb{R}^{(S+1)\times 1}$ of representing active sound sources:

$$P_{pred} = \mathbf{Sigmoid}(\mathbf{Linear}(\mathbf{a}_0, \mathbf{a}_1, \ldots, \mathbf{a}_s)). \tag{2}$$

During training, the target labels of sound existence are defined as $P_{gt} = [p_{1,\ldots,s} = 1,\ p_{s+1} = 0]$, with s being the ground-truth number of sources, and optimized using binary cross-entropy (BCE):

$$\mathcal{L}_p = \mathbf{BCE}(P_{pred}, P_{gt}). \tag{3}$$

At inference, the decoding loop (1) is repeated until the probability of the $(s+1)$-th source falls below a predefined threshold $\tau = 0.5$.

2.3. Multi-Task Learning Module

In sound source localization, existing deep learning network methods often treat envelope estimation and DOA prediction as separate and independent tasks. However, neurophysiological studies indicate that the human auditory system combines temporal and spatial cues to efficiently localize sounds and form perceptual attractors[18][19][20]. Motivated by this mechanism, we propose an envelope separation aided (ESA) method within a multi-task learning (MTL) model. As illustrated in Fig. 3, the Multi-Task Learning Module comprises two branches: one for envelope separation and another for source localization. By jointly optimizing these tasks through a gating mechanism, the multi-task learning model integrates temporal and spatial cues to enhance multi-source localization performance and enable mutual assistance between the envelope separation and direction prediction.

2.3.1. Envelope Separation Branch

The envelope separation branch is designed to separate the temporal envelope of each sound source from the mixed attractor embeddings. The output acoustic features $\mathbf{B}_{mix}$ from AFEM is concatenated and reshaped to $\mathbf{B}_{mix'} \in \mathbb{R}^{F\times 1\times T\times 2H}$. Meanwhile, adaptive attractor vectors $[\mathbf{a}_0, \mathbf{a}_1, \ldots, \mathbf{a}_{S-1}] \in \mathbb{R}^{F\times S\times 2H}$ (S is the number of sources) generated by the AAM are reshaped to $A \in \mathbb{R}^{F\times S\times 1\times 2H}$. Multiplying $\mathbf{A}$ with $\mathbf{B}_{mix'}$ produces the source-specific attractor embedding features $\mathbf{B}_A \in \mathbb{R}^{F\times S\times T\times 2H}$, which serve as input to the multi-task model.

To balance temporal and spatial information, a trainable gate vector $\mathbf{gate} \in \mathbb{R}^{1\times 1\times 2H}$ is introduced and constrained by a sigmoid activation, the gate is optimized during the initial n training epochs (we set $n = 3$) and then fixed to its learned value. The embedding features $\mathbf{B}_A$ are multiplied by $\mathbf{gate}$ to get $\mathbf{x}_{\text{env}}$ as input

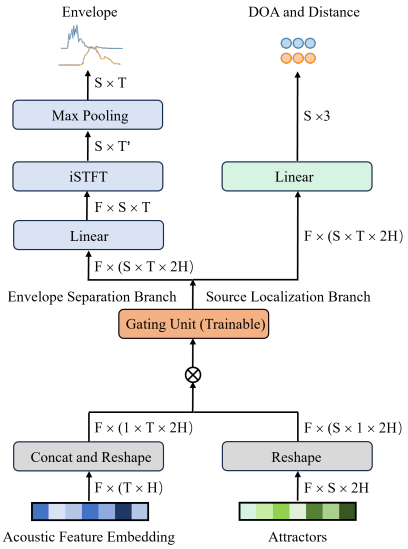


Fig. 3: The structure of Multi-Task Learning Module.

of a decoder of envelope estimation which is composed of a fully connected layer, an iSTFT layer and max-pooling layer to get the estimated envelopes.

$$\hat{e} = \textbf{MaxPool1d}(\textbf{iSTFT}(\textbf{Linear}(\mathbf{x}_{\text{env}}))) \in \mathbb{R}^{S\times T}, \quad (4)$$

2.3.2. Source Localization Branch

The source localization branch uses the same input attractor embedding $\mathbf{B}_A$ as the envelope branch, but multiplied by the complementary gate ($\mathbf{1}-\mathbf{gate}$) to get the embedding $\mathbf{x}_{xyz}$ to emphasize spatial cues, and then are reshaped to generate $\mathbf{x}_{\text{xyz}} \in \mathbb{R}^{S\times T\times 2FH}$ which are fed into a fully connected layer to regress 3D coordinates:

$$\hat{d} = \textbf{Linear}(\mathbf{x}_{\text{xyz}}) \in \mathbb{R}^{S\times T\times 3}, \quad (5)$$

where the last dimension corresponds to Cartesian coordinates (x, y, z), and the distance to each source is computed as the Euclidean norm of the coordinate vector.

2.3.3. Envelope Separation Aided Method

In multi-source scenarios, permutation issues can arise when the order of speakers in the ground truth does not match the order of outputs [13]. To overcome this, we adopt a joint optimization strategy by PIT [10] that aligns the envelope separation and direction prediction by minimizing their joint loss $\mathcal{L}_j$. The optimal permutation with the lowest joint loss is used for calculating the total loss $\mathcal{L}$ for backward propagation. For all source permutations $\pi \in \mathcal{P}_S$, and task-balancing weight λ (set $\lambda = 0.01$ in our experiments), the joint loss is defined as:

$$\mathcal{L}_j = \min_{\pi\in\mathcal{P}_S}\Big(\sum_{i=1}^{S}(\lambda\mathcal{L}_{env}(\hat{\mathbf{e}}_{\pi(i)} - \mathbf{e}_i) + \mathcal{L}_{xyz}(\hat{\mathbf{d}}_{\pi(i)} - \mathbf{d}_i))\Big) \quad (6)$$

where $\{\mathbf{e}_i\}_{i=1}^{S}$ and $\{\hat{\mathbf{e}}_j\}_{j=1}^{S}$ denote the ground-truth and predicted envelopes, while $\{\mathbf{d}_i\}_{i=1}^{S}$ and $\{\hat{\mathbf{d}}_j\}_{j=1}^{S}$ denote the corresponding spatial coordinates, $\mathcal{L}_{env}$ and $\mathcal{L}_{xyz}$ denote the envelope and coordinate losses respectively, with a numerical stability constant $\varepsilon = 10^{-10}$:

$$\mathcal{L}_{env}(\hat{\mathbf{e}}, \mathbf{e}) = \frac{\|\hat{\mathbf{e}} - \mathbf{e}\|_2^2}{\|\mathbf{e}\|_2^2 + \varepsilon} \quad (7)$$

$$\mathcal{L}_{xyz}(\hat{\mathbf{d}}, \mathbf{d}) = \|\hat{\mathbf{d}} - \mathbf{d}\|_2^2 \quad (8)$$

This joint PIT approach solves the permutation ambiguity problem and enables joint optimization of envelope and direction predictions, thereby enhancing localization accuracy. The overall training objective is:

$$\mathcal{L} = \mathcal{L}_p + \alpha\mathcal{L}_j. \quad (9)$$

where α controls the contribution of the joint loss term (Here we choose $\alpha = 1$).

3. EXPERIMENTS

3.1. Experimental Setup

We used simulated FOA signals generated from the FSD50K dataset [22] for training, testing, and evaluation. Room impulse responses (RIRs) were generated with the gpuRIR toolbox [23] using the image-source method [24]. Room dimensions were uniformly sampled from $[3\times3\times3]$ m^3 to $[10\times10\times6]$ m^3, and reverberation times (T_{60}) ranged from 0.2 to 1.0 s. The absorption coefficients of the room surfaces were randomly sampled between 0.5 and 1.0. The microphone array configured as an Eigenmike microphone array [25] was randomly located within 10% of the room size from the room center in each dimension. The number of sound sources was randomly selected from 1 to 3. We generated 20,000 samples for training and 4,000 samples for testing and evaluation. The network was trained using the Adam optimizer [26], with a learning rate of 1×10^{-4}, a batch size of 2. The maximum number of epochs is set to 200. The loss function is the sum of the BCE loss of the number of sound sources and the loss of the joint optimization with PIT shown in Eq. 9

We compared the proposed Envelope Separation Aided multi-task learning model with several baselines, including EINV2 [9], ACCDOA-trained SpatialNet (Sp-ACCDOA) [12], and SEET [17]. In addition, we conducted ablation experiments, further comparing the single-task direction coordinates regression without **E**nvelope **S**eparation **A**ided (**w/o ESA**) and the proposed **E**nvelope **S**eparation **A**ided multi-task learning (**w/ ESA**) model under different source number conditions.

3.2. Results

3.2.1. Source Number Prediction

The blind source counting accuracy (acc) is calculated as:

$$acc = \frac{\sum \mathbb{I}(\hat{\mathbf{S}} = \mathbf{S})}{M}, \quad (10)$$

where M denotes the total number of samples, and $\mathbf{S}$ and $\hat{\mathbf{S}}$ represent the ground-truth and predicted number of sound sources, respectively. Table 1 summarizes the source counting accuracy of our method (w/ ESA) and the baseline methods under different numbers of sources. We also reports the average accuracy ("Avg" in table), where the number of sources is randomly and uniformly selected from 1 to 3. In addition, we conducted ablation studies, confirming that the proposed Envelope Eeparation Aided Multi-Task Learning method consistently improves blind source counting accuracy.

Table 1: Results on Blind Source Counting Accuracy(%).

Model	Sources			
	1	**2**	**3**	**Avg**
EINV2	97.7	2.74	100.0	66.4
Sp-ACCDOA	100.0	63.1	98.1	87.3
SEET	92.6	90.5	80.3	88.0
ours (w/o ESA)	96.0	80.5	86.2	87.8
ours (w/ ESA)	**96.9**	**90.9**	**91.9**	**93.4**

3.2.2. DOA Estimation

Localization performance is evaluated using direction prediction errors, computed separately for azimuth, elevation, and distance for each source. The azimuth error (ε_{azi}) is defined as the minimum absolute difference between predicted and ground-truth azimuths ($\leq 180^\circ$), the elevation error (ε_{ele}) is the absolute difference in elevation, and the distance error is the absolute difference between predicted and ground-truth distances. We also calculate the relative distance error:

$$\varepsilon_{\text{rel}} = \frac{1}{S}\sum_{i=1}^{S}\frac{|\hat{l}_i - l_i|}{l_i + \varepsilon}, \tag{11}$$

where l and $\hat{l}$ are respectively the ground truth and predicted distance, S is the number of sound sources, $\varepsilon = 10^{-10}$ is a numerical stability constant. Table 2 summarizes localization errors for different methods including comparison and ablation experiments. Our proposed Envelope Separation Aided multi-task learning achieves the lowest errors across all metrics. Moreover, compared with single-task direction regression (w/o ESA), our multi-task learning model significantly improves blind source counting and localization accuracy, validating the effectiveness of our proposed model.

Table 2: Results on Blind Source Localization Errors.

Model	ε_{azi} (°)	ε_{ele} (°)	ε_{dist}(m)	ε_{rel}(%)
EINV2	86.07	35.97	2.47	62.41
Sp-ACCDOA	21.79	12.79	1.01	34.90
SEET	17.52	10.32	0.82	25.68
ours (w/o ESA)	16.31	10.09	0.75	26.69
ours (w/ ESA)	**10.59**	**6.74**	**0.64**	**22.08**

3.2.3. Discussion

We further present heatmaps of the DOA estimates and source number counting predictions performance, under the average case where the number of sources is randomly and uniformly selected from 1 to 3 for different methods with a 5° grid for azimuth and elevation and a $0.5m$ grid for distance, as shown in Fig. 4. For blind source counting, the heatmaps reveal that our multi-task learning model substantially enhances the accuracy of source count prediction when two sources are present. In contrast, the single-task model performs noticeably worse in two-source scenarios compared to cases with one or three sources. For blind source localization, Our model produces sharper and more concentrated peaks around the true source locations. Moreover, the uniform distribution of the data across diverse azimuths, elevations, and distances further demonstrates that our method achieves superior accuracy and robustness in multi-source localization.

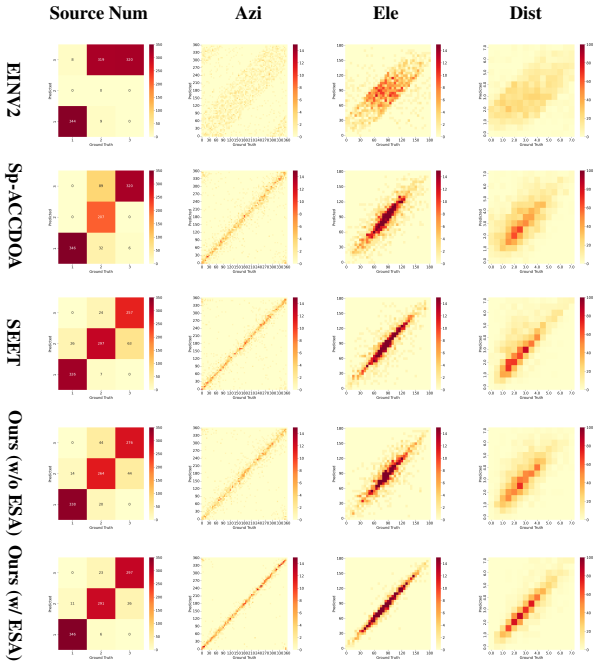


Fig. 4: Heatmaps of source number counting and localization results for 1-3 sources

In addition, we applied t-SNE [27] to visualize the high-dimensional attractors. Fig. 5 shows the attractor distributions for 10,000 sound mixture samples, with the number of sources ranging from 2 to 4. As shown in Fig. 5, attractors corresponding to different sources are clearly well separated in the embedding space, demonstrating the effectiveness of the Adaptive Attractor Module.

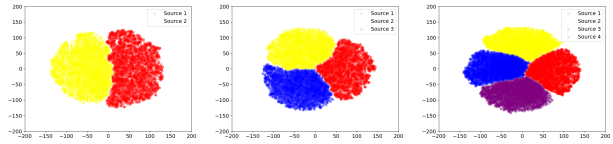


Fig. 5: T-SNE results of attractors in embedding space under different numbers of sources.

4. CONCLUSION

In this paper, we proposed an **E**nvelope **S**eparation **A**ided **M**ulti-**T**sak **L**earning model. By jointly optimizing temporal-domain envelope separation and spatial coordinates and distance estimation through a unified loss with PIT, our model effectively leverages mutual assistance between temporal and spatial cues. Experiments demonstrated that envelope separation provides complementary temporal cues that enhance direction estimation, enabling more accurate and robust blind source counting and localization. Future work includes integrating more advanced architectures, such as Transformer-based decoders and U-Net structures, to further strengthen the synergy between temporal envelope features and spatial direction information.

5. REFERENCES

[1] Pierre-Amaury Grumiaux, Srđan Kitić, Laurent Girin, and Alexandre Guérin, "A survey of sound source localization with deep learning methods," *The Journal of the Acoustical Society of America*, vol. 152, no. 1, pp. 107–151, 2022.

[2] Xinya Li, Zhiqun Daniel Deng, Lynn T Rauchenstein, and Thomas J Carlson, "Contributed review: Source-localization algorithms and applications using time of arrival and time difference of arrival measurements," *Review of Scientific Instruments*, vol. 87, no. 4, 2016.

[3] Ralph Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE transactions on antennas and propagation*, vol. 34, no. 3, pp. 276–280, 1986.

[4] Joe C Chen, Kung Yao, and Ralph E Hudson, "Source localization and beamforming," *IEEE Signal Processing Magazine*, vol. 19, no. 2, pp. 30–39, 2002.

[5] Kunbo Xu, Zekai Zong, Dongjun Liu, Ran Wang, and Liang Yu, "Deep learning-based sound source localization: A review," *Applied Sciences*, vol. 15, no. 13, pp. 7419, 2025.

[6] John R Hershey, Zhuo Chen, Jonathan Le Roux, and Shinji Watanabe, "Deep clustering: Discriminative embeddings for segmentation and separation," in *2016 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2016, pp. 31–35.

[7] Lauréline Perotin, Alexandre Défossez, Emmanuel Vincent, Romain Serizel, and Alexandre Guérin, "Regression versus classification for neural network based audio source localization," in *2019 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2019, pp. 343–347.

[8] Sharath Adavanne, Archontis Politis, Joonas Nikunen, and Tuomas Virtanen, "Sound event localization and detection of overlapping sources using convolutional recurrent neural networks," *IEEE Journal of Selected Topics in Signal Processing*, vol. 13, no. 1, pp. 34–48, 2018.

[9] Jinbo Hu, Yin Cao, Ming Wu, Qiuqiang Kong, Feiran Yang, Mark D Plumbley, and Jun Yang, "A track-wise ensemble event independent network for polyphonic sound event localization and detection," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 9196–9200.

[10] Morten Kolbæk, Dong Yu, Zheng-Hua Tan, and Jesper Jensen, "Multitalker speech separation with utterance-level permutation invariant training of deep recurrent neural networks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 25, no. 10, pp. 1901–1913, 2017.

[11] Kazuki Shimada, Yuichiro Koyama, Naoya Takahashi, Shusuke Takahashi, and Yuki Mitsufuji, "Accdoa: Activity-coupled cartesian direction of arrival representation for sound event localization and detection," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 915–919.

[12] Kazuki Shimada, Yuichiro Koyama, Shusuke Takahashi, Naoya Takahashi, Emiru Tsunoo, and Yuki Mitsufuji, "Multi-accdoa: Localizing and detecting overlapping sounds from the same class with auxiliary duplicating permutation invariant training," in *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2022, pp. 316–320.

[13] Yi Luo, Zhuo Chen, and Nima Mesgarani, "Speaker-independent speech separation with deep attractor network," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 26, no. 4, pp. 787–796, 2018.

[14] Zhengyang Chen, Bing Han, Shuai Wang, and Yanmin Qian, "Attention-based encoder-decoder end-to-end neural diarization with embedding enhancer," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 1636–1649, 2024.

[15] Srikanth Raj Chetupalli and Emanuël AP Habets, "Speech separation for an unknown number of speakers using transformers with encoder-decoder attractors.," in *Interspeech*, 2022, pp. 5393–5397.

[16] Younglo Lee, Shukjae Choi, Byeong-Yeol Kim, Zhong-Qiu Wang, and Shinji Watanabe, "Boosting unknown-number speaker separation with transformer decoder-based attractor," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 446–450.

[17] Donghang Wu, Jiaqi Du, Tianshu Qu, Qingbo Huang, and Dejun Zhang, "Moving sound source localization and tracking based on envelope estimation for unknown number of sources," 05 2025.

[18] Patricia K Kuhl, "Human adults and human infants show a perceptual magnet effect for the prototypes of speech categories, monkeys do not," *Perception & psychophysics*, vol. 50, no. 2, pp. 93–107, 1991.

[19] John C Middlebrooks, "Sound localization," *Handbook of clinical neurology*, vol. 129, pp. 99–116, 2015.

[20] Jens Blauert, *Spatial hearing: the psychophysics of human sound localization*, MIT press, 1997.

[21] Changsheng Quan and Xiaofei Li, "Spatialnet: Extensively learning spatial information for multichannel joint speech separation, denoising and dereverberation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 1310–1323, 2024.

[22] Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra, "Fsd50k: An open dataset of human-labeled sound events," 2022.

[23] David Diaz-Guerra, Antonio Miguel, and Jose R Beltran, "gpurir: A python library for room impulse response simulation with gpu acceleration," *Multimedia Tools and Applications*, vol. 80, no. 4, pp. 5653–5671, 2021.

[24] Jont B Allen and David A Berkley, "Image method for efficiently simulating small-room acoustics," *The Journal of the Acoustical Society of America*, vol. 65, no. 4, pp. 943–950, 1979.

[25] Edwin Pfanzagl-Cardone, "Hoahigher order ambisonics (eigenmike®)," in *The Art and Science of 3D Audio Recording*, pp. 189–209. Springer, 2023.

[26] Diederik P Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[27] Laurens van der Maaten and Geoffrey Hinton, "Visualizing data using t-sne," *Journal of machine learning research*, vol. 9, no. Nov, pp. 2579–2605, 2008.